

Troubleshooting Guide

Troubleshooting Guide

Common Issues and Solutions

Service Won't Start

Symptoms

- `./start` command fails
- Containers exit immediately
- Error messages in logs

Diagnosis

```
# Check container runtime
./init --check
```

```
# View logs
./logs/proxy.log
podman logs proxy-squid
```

Solutions

1. Port already in use

```
# Find process using port
ss -tuln | grep 53128
# Kill process or change port in .env
```

2. Permission denied

```
# Fix permissions
chmod +x init start stop status cache
```

3. Missing dependencies

```
# Install Docker or Podman
# Ubuntu/Debian
```

```
sudo apt install docker.io docker-compose
```

```
# Fedora
```

```
sudo dnf install podman podman-compose
```

4. Configuration errors

```
# Validate .env
```

```
./init --check
```

VPN Not Connecting

Symptoms

- VPN container exits
- IP not masked
- Connection timeouts

Diagnosis

```
# Check VPN container status
```

```
podman ps -a | grep vpn
```

```
# View VPN logs
```

```
podman logs proxy-vpn
```

```
# Check configuration
```

```
cat .env | grep VPN
```

Solutions

1. Invalid credentials

```
# Verify in .env
```

```
VPN_USERNAME=correct_username
```

```
VPN_PASSWORD=correct_password
```

2. Missing .ovpn file

```
# Check file exists
```

```
ls -la $VPN_OVPN_PATH
```

```
# Update path in .env
```

```
VPN_OVPN_PATH=/correct/path/to/config.ovpn
```

3. Permission issues

```
# Check .ovpn permissions
```

```
chmod 644 $VPN_OVPN_PATH
```

4. Network issues

```
# Test VPN server connectivity
ping $(grep remote config.ovpn | awk '{print $2}')
```

Cache Not Working

Symptoms

- No cache growth
- High bandwidth usage
- Slow repeated requests

Diagnosis

```
# Check cache stats
./cachectl stats

# Check Squid logs
tail -f logs/squid/cache.log

# Verify cache directory
ls -la $CACHE_DIR/squid
```

Solutions

1. Cache directory not writable

```
# Fix permissions
chmod -R 755 $CACHE_DIR
chown -R $PUID:$PGID $CACHE_DIR
```

2. Cache full

```
# Clear cache
./cachectl clear

# Or increase limit
CACHE_MAX_SIZE_GB=100
```

3. Squid configuration error

```
# Test config
podman exec proxy-squid squid -k parse

# Rebuild config
./init
```

4. Objects not cacheable

- Check Cache-Control headers
 - Review refresh patterns
 - Enable HTTPS inspection for HTTPS caching
-

Connection Refused

Symptoms

- "Connection refused" errors
- Timeout when connecting
- Proxy not responding

Diagnosis

Check if service running

```
./status
```

Check port binding

```
ss -tuln | grep -E '53128|51080'
```

Test local connection

```
curl --proxy http://localhost:53128 http://example.com
```

Solutions

1. Service not running

```
./start
```

2. Wrong port

Check configured port

```
cat .env | grep PORT
```

3. Firewall blocking

Allow ports

```
sudo ufw allow 53128/tcp
```

```
sudo ufw allow 51080/tcp
```

4. Bind address wrong

Use 0.0.0.0 for all interfaces

```
BIND_ADDRESS=0.0.0.0
```

DNS Resolution Fails

Symptoms

- "DNS resolution failed" errors
- Can ping IPs but not domains
- Host not found errors

Diagnosis

Test DNS

```
nslookup google.com
```

Check DNS config

```
cat /etc/resolv.conf
```

Test via proxy

```
curl --proxy http://localhost:53128 http://google.com
```

Solutions

1. Configure DNS servers

```
DNS_SERVERS=8.8.8.8,8.8.4.4
```

2. Enable DNS caching

```
DNS_CACHING_ENABLED=true
```

```
DNS_CACHE_TTL=3600
```

3. VPN DNS issues

- Check .ovpn file for DNS settings
- Add dhcp-option DNS 8.8.8.8 to .ovpn

High Memory Usage

Symptoms

- System running low on memory
- OOM kills
- Slow performance

Diagnosis

Check container memory

```
podman stats
```

```
# Check cache memory
./cachectl stats
```

Solutions

1. Reduce cache memory

```
CACHE_MEMORY_SIZE_MB=256
```

2. Limit connections

```
MAX_CONNECTIONS=1024
```

3. Reduce cache size

```
./cachectl trim 20
```

Slow Performance

Symptoms

- Slow page loads
- High latency
- Timeouts

Diagnosis

```
# Check hit ratio
./cachectl stats
```

```
# Monitor connections
podman exec proxy-squid squidclient mgr:info
```

```
# Check network
ping -c 10 8.8.8.8
```

Solutions

1. Increase cache memory

```
CACHE_MEMORY_SIZE_MB=1024
```

2. Optimize refresh patterns

- Edit config/squid/squid.conf
- Add domain-specific patterns

3. Check VPN latency

- Try different VPN server

- Use closer server location

4. Network issues

- Check bandwidth
 - Look for network congestion
-

VPN Disconnects Frequently

Symptoms

- VPN connection drops
- IP changes unexpectedly
- Connection interruptions

Diagnosis

```
# Check VPN status
./status
```

```
# View VPN logs
podman logs proxy-vpn
```

```
# Monitor continuously
./status --watch
```

Solutions

1. Enable auto-reconnect

```
VPN_AUTO_RECONNECT=true
```

2. Add keepalive

```
# Add to .ovpn file
keepalive 10 60
```

3. Switch servers

- Try different VPN server
- Use more reliable provider

4. Check network stability

- Test without VPN
 - Check ISP issues
-

Permission Denied Errors

Symptoms

- "Permission denied" in logs
- Cannot write to cache
- Cannot read config files

Diagnosis

```
# Check permissions
ls -la config/
ls -la cache/
ls -la logs/

# Check user/group
id
```

Solutions

1. Fix directory permissions

```
chmod -R 755 config cache logs
chown -R $PUID:$PGID config cache logs
```

2. Fix file permissions

```
chmod 644 config/squid/squid.conf
chmod 600 vpn-auth.txt
```

3. Correct PUID/PGID

```
# Get your IDs
id

# Set in .env
PUID=1000
PGID=1000
```

Container Crashes

Symptoms

- Containers exit unexpectedly
- Restart loops
- Service unavailable

Diagnosis

Check container status

```
podman ps -a
```

View crash logs

```
podman logs --tail 100 proxy-squid
```

```
podman logs --tail 100 proxy-vpn
```

Solutions

1. Resource limits

Check system resources

```
free -h
```

```
df -h
```

2. Configuration error

Rebuild config

```
./init --force
```

3. Image issues

Pull fresh images

```
./start --pull
```

4. Kernel issues (Podman)

Check kernel version

```
uname -r
```

Need 4.18+ for rootless Podman

Diagnostic Commands

Quick Diagnostics

Full system check

```
./status -v
```

Configuration check

```
./init --check
```

Cache health

```
./cachectl stats
```

Network test

```
curl --proxy http://localhost:53128 https://ifconfig.me
```

Log Collection

```
# Collect all logs
mkdir -p /tmp/proxy-logs
cp -r logs/* /tmp/proxy-logs/
podman logs proxy-squid > /tmp/proxy-logs/squid.log 2>&1
podman logs proxy-dante > /tmp/proxy-logs/dante.log 2>&1
podman logs proxy-vpn > /tmp/proxy-logs/vpn.log 2>&1

# Create archive
tar -czvf proxy-logs.tar.gz -C /tmp proxy-logs
```

Network Diagnostics

```
# Check ports
ss -tuln | grep -E '53128|51080|58080'

# Test connectivity
ping -c 5 8.8.8.8
traceroute 8.8.8.8

# DNS test
dig google.com
nslookup google.com

# Proxy test
curl -v --proxy http://localhost:53128 http://example.com
```

Getting Help

1. **Check documentation:** See docs/ directory
2. **Search issues:** GitHub Issues
3. **Collect diagnostics:** Use log collection above
4. **Open issue:** Include diagnostics and error messages